Ensemble Machine Learning Approaches for Predicting Academic Distress and Mental Health Risk in Higher Education: A Multi-Dataset Comparative Study

Rita Chineyen Nicholas
Ulster University / Computer Science with Technology and Business Development
ID: 20018531 | Birmingham, United Kingdom

Abstract

Student mental health challenges, including stress and depression, continue to pose significant concerns within higher education globally, affecting academic performance, retention, and overall wellbeing. Despite the increasing availability of student-related data, institutions often lack predictive systems capable of identifying at-risk individuals early enough for effective intervention. This study explores the application of machine learning techniques to predict academic distress and mental health risk using multi-dimensional student data drawn from publicly available international datasets. A comparative approach is adopted, utilising two distinct datasets focusing on stress and depression respectively. Multiple machine learning models, including Logistic Regression, Decision Tree, and Random Forest, are implemented and evaluated. The findings indicate that model performance varies based on data characteristics, with simpler models demonstrating strong predictive capability in structured datasets. Robustness testing further confirms that the models maintain stable performance under varying data conditions. The study contributes to educational data mining by providing a robust, interpretable, and data-driven framework for early identification of at-risk students, with broader implications for higher education institutions worldwide.

Keywords—Machine Learning, Student Mental Health, Stress Prediction, Depression Prediction, Educational Data Mining, Logistic Regression, Random Forest

I. INTRODUCTION

Mental health issues among university students have become a global concern, with increasing levels of stress, anxiety, and depression affecting academic outcomes and overall well-being [14], [15]. Many students experiencing mental health challenges do not seek help, resulting in delayed intervention and adverse consequences.

Higher education institutions collect extensive student data, including academic performance, behavioural patterns, and lifestyle indicators. However, this data is often underutilised in identifying at-risk students. Machine learning offers an opportunity to transform such data into predictive systems capable of enabling early intervention [8], [9].

Limited studies compare multiple machine learning models across distinct mental health datasets within a unified framework. This study addresses this gap by evaluating model performance across both stress and depression datasets. While the datasets employed are drawn from international sources, the findings carry particular relevance for higher education institutions globally, including in the United Kingdom, where student mental health has emerged as a growing institutional concern [15].

II. RELATED WORK

Previous research has demonstrated the potential of machine learning in predicting student outcomes. Studies have shown that mental health challenges significantly impact academic performance and retention [14].

Machine learning techniques such as decision trees, neural networks, and ensemble methods have been widely applied in predicting psychological outcomes [8], [11]. Psychological variables such as anxiety and depression consistently emerge as strong predictors.

Recent studies have highlighted the role of digital behaviour in influencing mental health. Excessive internet use and digital engagement have been linked to increased psychological distress [1], [2], [3].

However, publicly available datasets often lack behavioural indicators such as screen time and social media usage, limiting model comprehensiveness. Additionally, limited research compares the effectiveness of simple and complex models across multiple datasets, which this study aims to address.

III. METHODOLOGY

A. Research Design

This study adopts a quantitative, data-driven approach using supervised machine learning techniques. A multi-dataset comparative design was employed:

- Dataset A: Depression prediction (binary classification)
- Dataset B: Stress prediction (multiclass classification)

This approach enables cross-validation of predictive performance across different mental health outcomes [8].

B. Data Description and Dataset Selection

The datasets [16], [17] consist of student-related variables spanning psychological, academic, physiological, and socio-environmental dimensions. The depression dataset focuses on binary classification, while the stress dataset includes multiple ordinal categories representing varying stress levels. Both datasets are structured and contain a mix of categorical and numerical features suitable for supervised learning.

- Dataset A — Depression dataset (target: 0 = No, 1 = Yes)
- Dataset B — Stress dataset (target: 0 = Low, 1 = Medium, 2 = High)

The datasets were selected based on their relevance to the research objective and their suitability for supervised machine learning. Both contain a diverse range of student-related variables which are critical for modelling mental health outcomes. The datasets are structured and sufficiently large for training and evaluating classification models, with clearly defined target variables. Public availability via Kaggle ensures reproducibility and transparency. While alternative datasets exist, many lack the combination of psychological and contextual variables required for comprehensive analysis, making these datasets particularly suitable.

Table I presents all variables included in Dataset A (depression), along with a description of each. The variable 'Depression' serves as the binary target for classification.

TABLE I. VARIABLE DESCRIPTIONS — DATASET A (DEPRESSION)

Variable	Description
Depression	Target variable indicating presence of depression (0 = No, 1 = Yes)
Gender	Gender of the student respondent
Age	Age of the student respondent
City	City of residence (grouped into categories)
Profession	Student's current profession or role
Academic Pressure	Level of perceived academic pressure experienced
Work Pressure	Level of work-related pressure experienced
CGPA	Cumulative Grade Point Average, discretised into bands
Study Satisfaction	Student's satisfaction with their academic experience
Job Satisfaction	Level of satisfaction with current employment
Sleep Duration	Average hours of sleep per night reported
Dietary Habits	Quality and regularity of diet reported by student
Degree Field	Field of academic study (grouped into categories)
Suicidal Thoughts	Indicator of whether student has experienced suicidal ideation
Work/Study Hours	Average daily hours spent on work or study
Financial Stress	Degree of financial difficulty or concern reported
FM Mental Illness	Family history of mental illness (binary indicator)

Table II presents all variables included in Dataset B (stress), along with a description of each. The variable 'stress_level' serves as the multiclass target for classification.

TABLE II. VARIABLE DESCRIPTIONS — DATASET B (STRESS)

Variable	Description
stress_level	Target variable: stress category (0 = Low, 1 = Medium, 2 = High)
anxiety_level	Self-reported level of anxiety experienced
self_esteem	Student's reported level of self-esteem
mental_health_history	Prior history of mental health conditions
depression	Indicator of depressive symptoms
headache	Frequency of headaches reported
blood_pressure	Reported blood pressure level
sleep_quality	Quality of sleep experienced
breathing_problem	Presence of breathing difficulties
noise_level	Exposure to disruptive noise in environment
living_conditions	Quality of student's living environment
safety	Perceived personal safety level
basic_needs	Extent to which basic needs are met
academic_performance	Student academic performance level
study_load	Academic workload experienced by student
teacher_student_relationship	Quality of relationship with academic staff
future_career_concerns	Level of concern about future career prospects
social_support	Availability of emotional/social support
peer_pressure	Influence from peers affecting stress
extracurricular_activities	Level of participation in extracurricular activities
bullying	Exposure to bullying experiences

Fig. 1 shows the class distribution of Dataset A, illustrating the proportion of students classified as depressed versus non-depressed across the full sample.

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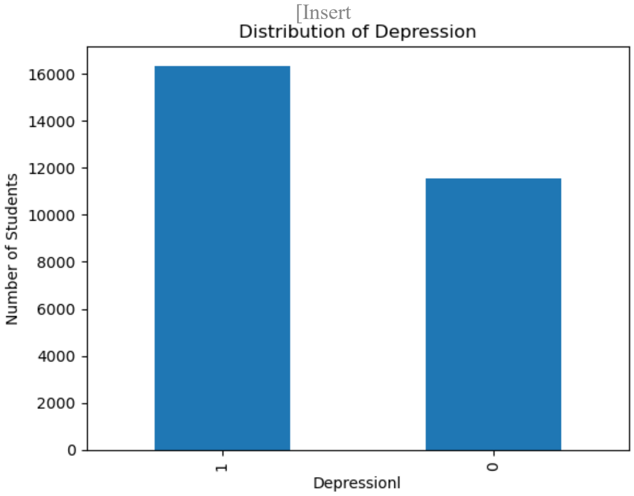
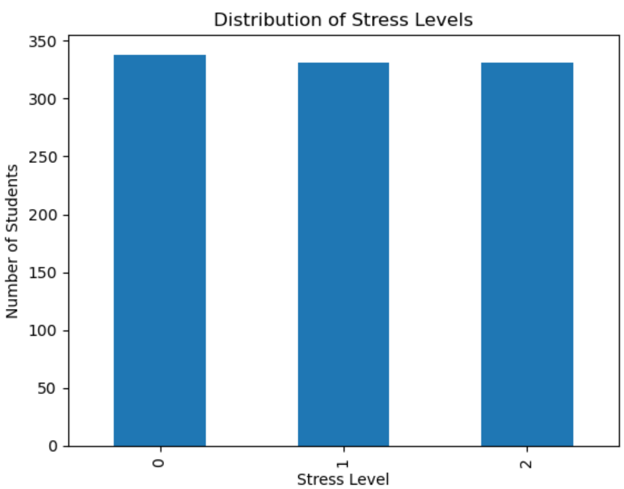


Fig. 2 shows the class distribution of Dataset B, illustrating the proportion of students falling into low, medium, and high stress categories.



C. Data Preprocessing and Feature Engineering

Data preprocessing was a critical step in ensuring model performance and reliability. Irrelevant features such as identifiers and raw categorical labels were removed to reduce noise. Categorical variables were grouped and encoded for compatibility with machine learning algorithms, while continuous variables such as CGPA and age were discretised to capture meaningful ranges and improve interpretability. Mode imputation was applied to handle missing values, as it is well-suited for categorical datasets and preserves the most frequent patterns [6]. These decisions were guided by established data mining practices balancing performance with interpretability [5], [10].

D. Models Used

Logistic Regression was selected for its effectiveness with structured datasets and its interpretability, which is particularly important in educational and mental health contexts [6]. Decision Tree and Random Forest models were included to capture potential non-linear relationships and interactions between variables. Random Forest, as an ensemble method, is known for robustness and reduced overfitting [4]. The inclusion of multiple models allows comprehensive evaluation and ensures the selected model fits the underlying data characteristics [8].

E. Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, and F1-Score. These metrics provide a comprehensive assessment of classification performance, capturing both false positive and false negative rates [5].

F. Model Robustness Testing

To ensure model robustness, key edge cases were evaluated, including missing data, outlier sensitivity, and multicollinearity. These were selected due to their direct relevance to real-world data challenges and their impact on model performance and interpretability.

G. Implementation Environment

The study was implemented using Python (Jupyter Notebook) with the following libraries:

- Pandas, NumPy — data manipulation and numerical processing
- Scikit-learn — model training, evaluation, and preprocessing
- Matplotlib, Seaborn — data visualisation and correlation analysis

IV. RESULTS

A. Stress Prediction

Logistic Regression achieved the highest accuracy (0.915), outperforming Decision Tree and Random Forest. This suggests that the relationships within the dataset may be predominantly linear, reducing the advantage of ensemble methods [6].

Key Insight: Stress is influenced by short-term psychological and academic factors such as anxiety, peer pressure, and study load.

Fig. 3 displays the top features contributing to stress level prediction, ranked by their correlation coefficient magnitude as derived from the trained models.

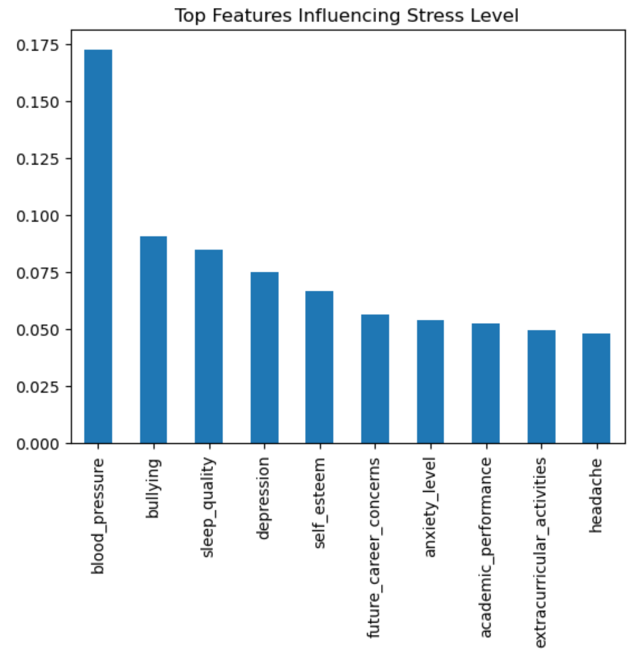
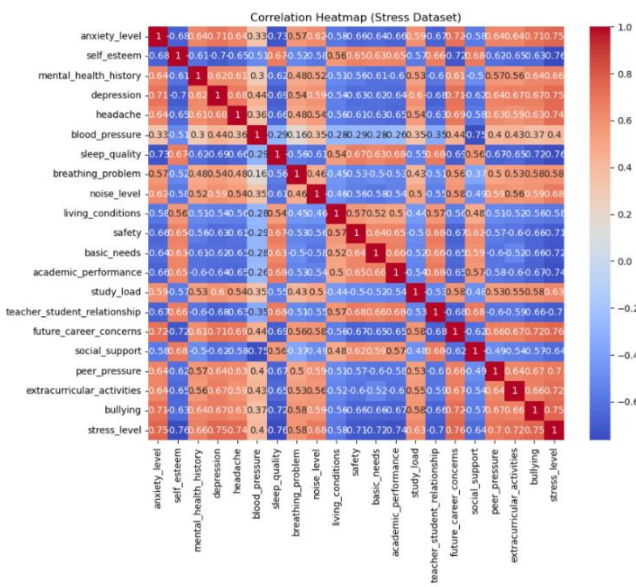


Fig. 4 presents a heatmap of pairwise correlations between all variables in Dataset B, illustrating the strength and direction of each relationship with stress level.



The correlation analysis reveals that stress levels are most strongly associated with future career concerns (0.7569), bullying (0.7532), anxiety (0.7530), and depression (0.7512), indicating that psychological and social stressors are the primary drivers. Strong positive relationships are also observed with headache (0.7390) and extracurricular activities (0.7223). In contrast, study load exhibits a moderate correlation (0.6320), indicating academic pressure is relevant but not dominant. Strong negative correlations are observed with sleep quality (−0.7646), self-esteem (−0.7607), and academic performance (−0.7378), highlighting the protective role of wellbeing and academic stability.

B. Depression Prediction

Depression prediction demonstrated a stronger influence from long-term factors such as financial stress and lifestyle conditions.

Key Insight: Depression reflects cumulative and persistent conditions, consistent with existing mental health research [14].

Fig. 5 displays the top features contributing to depression prediction, showing which variables most strongly differentiate depressed from non-depressed students in Dataset A.

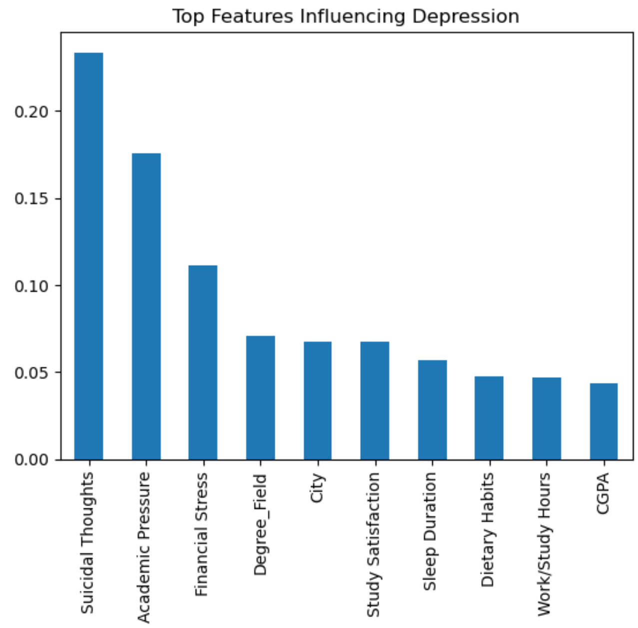
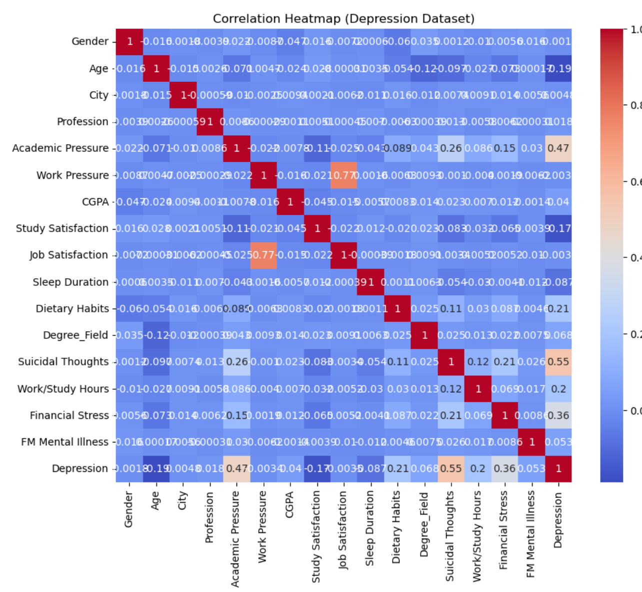


Fig. 6 presents a heatmap of pairwise correlations between all variables in Dataset A, illustrating the strength and direction of each relationship with the depression target variable.



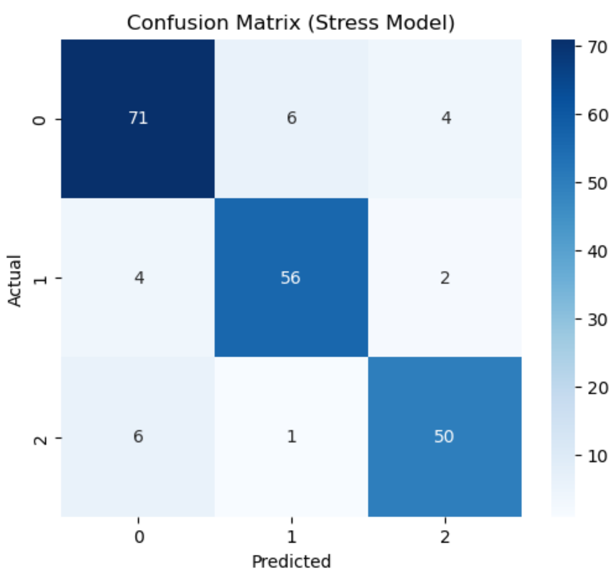
The correlation analysis for depression reveals no variables exhibiting strong correlations. The highest association is with suicidal thoughts (0.5463), followed by academic pressure

(0.4748) and financial stress (0.3635), indicating depression is influenced by a combination of psychological distress and external pressures rather than a single dominant factor. Weak negative correlations are observed with age (−0.1898), study satisfaction (−0.1680), and sleep duration (−0.0867). Overall, the findings highlight the complex and multifactorial nature of depression [14], making it less directly predictable compared to stress.

C. Robustness Results

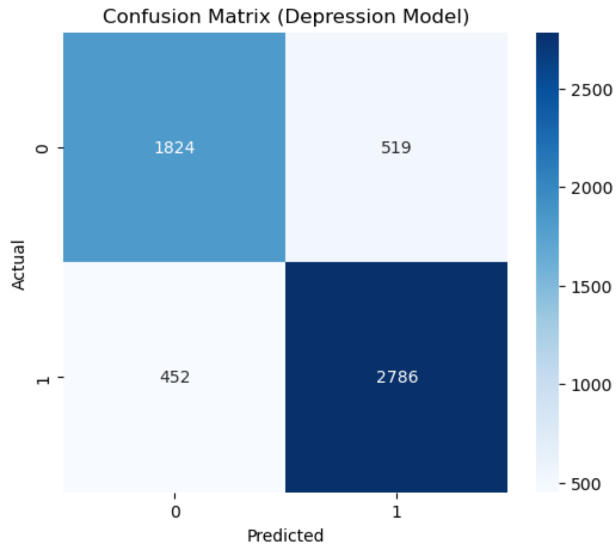
The robustness analysis indicates that the models maintain stable performance under varying data conditions. Mode imputation of missing values resulted in only a marginal reduction in accuracy, demonstrating resilience to incomplete data. The presence of outliers had negligible impact, particularly for Random Forest, which is inherently robust to extreme values. Multicollinearity analysis showed that removing highly correlated features did not significantly affect accuracy, confirming the models are not overly dependent on redundant variables.

Fig. 7 shows the confusion matrix for the stress prediction model, summarising true and predicted class counts across all three stress categories.



The stress confusion matrix shows strong classification performance across all three classes, with true positive values of 71 (low), 56 (medium), and 50 (high). Misclassifications occur primarily between adjacent stress levels, expected given the subjective and continuous nature of stress.

Fig. 8 shows the confusion matrix for the depression prediction model, summarising true positive, false positive, true negative, and false negative counts across both classes.



The depression confusion matrix indicates strong performance, with 2786 true positives. While 519 false positives are observed, the 452 false negatives remain relatively low — particularly important in mental health contexts where missed identifications carry serious consequences.

D. Model Comparative Analysis

Fig. 9 compares the performance of all three models on Dataset B across accuracy, precision, recall, and F1-score, providing a direct cross-model comparison for stress prediction.

	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.915	0.915643	0.915	0.914935
Decision Tree	0.890	0.891106	0.890	0.890156
Random Forest	0.885	0.885020	0.885	0.884972

Fig. 10 compares the performance of all three models on Dataset A across the same evaluation metrics, providing a direct cross-model comparison for depression prediction.

	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.833005	0.832451	0.833005	0.832454
Decision Tree	0.757212	0.757404	0.757212	0.757304
Random Forest	0.826017	0.825518	0.826017	0.825646

Logistic Regression achieved the highest performance across all metrics for both datasets. Decision Tree and Random Forest did not provide additional performance benefits, suggesting the datasets are well-structured and predominantly linear in nature. This reinforces the importance of selecting models based on data characteristics rather than assuming the superiority of more complex algorithms.

The balance between precision and recall is particularly important here. False negatives may leave at-risk students unidentified, while excessive false positives could trigger unnecessary intervention. Logistic Regression maintained strong performance across all metrics, indicating both high accuracy and balanced predictive capability.

V. DISCUSSION

Logistic Regression outperformed more complex models, indicating that the underlying relationships within the datasets may be predominantly linear. This aligns with established research suggesting that simpler models can perform effectively when data is well-structured and appropriately preprocessed [6]. The lack of superior performance from ensemble methods reinforces the importance of selecting models based on data nature rather than algorithmic assumptions [4].

Psychological variables, including anxiety and depression, were the most influential predictors of stress, supported by academic and social factors such as peer pressure and study load. These findings are consistent with existing literature [14]. Physiological factors such as sleep quality also showed a strong inverse relationship with stress levels, supporting a holistic understanding of student mental health.

Robustness testing confirmed stable model performance under missing values, outliers, and multicollinearity, demonstrating suitability for real-world deployment where data is rarely clean or complete.

The superior performance of Logistic Regression highlights the value of interpretability in sensitive domains such as mental health, where understanding contributing factors is as important as predictive accuracy [8]. While the datasets are international in scope, the findings are broadly applicable to higher education institutions globally, including those in the United Kingdom, where similar psychological and academic stressors are well documented [15].

VI. ETHICAL CONSIDERATIONS

The study utilised publicly available and anonymised datasets, ensuring that no personal identifiers were included. While predictive models may introduce risks of misclassification, they are intended to support, rather than replace, professional judgement in real-world applications.

VII. CONCLUSION

This study demonstrates that machine learning models can effectively predict student mental health outcomes across both stress and depression domains. Logistic Regression achieved the best performance across both datasets, highlighting the importance of data structure over model complexity. Key predictors include psychological, academic, social, and physiological factors, with sleep quality and anxiety emerging as particularly influential variables.

Robustness testing confirms model reliability under imperfect data conditions, supporting practical deployment in educational environments. The research question is answered affirmatively: machine learning models can effectively predict academic distress and mental health risk with high accuracy, with structured and predominantly linear data relationships identified as the primary driver of Logistic Regression’s outperformance.

The study is subject to several limitations. The datasets are international in scope and sourced from Kaggle, meaning findings may not directly reflect the characteristics of any single national context. The absence of behavioural data such as screen time limits model comprehensiveness, and self-reported variables introduce inherent bias. Overall generalisability remains limited to populations resembling the dataset demographics. Future work should explore institution-specific datasets, incorporate behavioural and longitudinal data, and investigate deployment within live educational management systems.

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